AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 16 · STUDENT LESSON

Regularisation and early stopping

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Separate the penalty. Replay the rule. Restore the saved model.

Trace regularized updates and strict-improvement stopping without conflating losses or inventing restoration evidence.

Keep one habit: Complete penalty terms → pictured separate exercises → full observed-prefix ledger → executable policy replay → independent new objective/log.

MAIN LESSON · SECTION 1

What changes the training objective,and what decides when to stop?

> Keep prediction loss,weight cost and checkpoint selection in separate records.

Learning goals

I can separate a **weight penalty** from **prediction** error.

I can trace a regularised update under a stated objective.

I can apply a stopping rule and **restore** the best observed checkpoint.

Retrieve saved-state validation selection

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from **Training** and validation curves. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- If validation chooses an earlier checkpoint, which parameters should be used after selection?

> **My starting explanation**

End goal

By the end, I can explain **regularisation** and early stopping using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Separate a **weight penalty** from **prediction** error

Trace a regularised update under a stated objective

Apply a stopping rule and **restore** the best observed checkpoint

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Add a declared cost to one specified weight

> The penalty changes the objective;it is not another prediction error.

Complete restricted model and penalty definition

Keep h = ReLU(x), y = vh, train only v on (1, 2) and (2, 4), and freeze all hidden parameters and the zero **output** bias. **Training** MSE is L(v) = 2.5(v - 2) squared.

Define J(v) = L(v) + lambda times v squared, with lambda nonnegative. Only this one **output** weight is penalised here.

For lambda = 0.5, the penalty at v = 1.5 is 0.5 times 2.25 = 1.125. **Prediction** MSE is 0.625, so total objective J is 1.75. Keep these quantities separate in logs.

A smaller total objective does not automatically mean a smaller **prediction** error.

Separate data loss,weight cost and total

v	Prediction MSE	Penalty at lambda 0.5	Total J
1	2.5	0.5	3
1.5	0.625	1.125	1.75

v	Prediction MSE	Penalty at lambda 0.5	Total J
2	0	2	2

Bound the three-candidate comparison

Among these three candidates, the smallest penalised objective is at v = 1.5, whereas the smallest **training** MSE is at v = 2. Penalising a weight changes the tradeoff.

It does not guarantee better validation performance or fair outcomes.

Name given and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Trace the data and penalty contributions into one update

> A change in total objective must not be mislabeled as validation prediction loss.

Complete gradient derivation

The data gradient is 5(v - 2). The penalty gradient is 2 times lambda times v, giving total gradient 5(v - 2) + 2 lambda v. At v = 1.5 and lambda = 0.5, these contributions are -2.5 and 1.5, so total gradient is -1.

A rate 0.1 step gives v = 1.6.

Two complete penalty-aware updates

Main lambda.5, rate.1	Start v1.5	Next start v1.6
Data gradient 5(v−2)	-2.5	-2
Penalty gradient 2 lambda v	1.5	1.6
Total gradient	-1	-0.4
New vafter one update	1.6	1.64
New prediction MSE	0.4	0.324
New penalty	1.28	1.3448
New total objective	1.68	1.6688

Objective convention and helper contract

Both gradient contributions are evaluated at the same old v. Keep the data term L=2.5(v−2)²and penalty lambda v²separate before adding them. The one-step code assumes finite numbers and nonnegative lambda; it is not a validator or a search for the best penalty. Comparing total Jvalues across different lambdas changes the measure being compared.

Use the same declared validation prediction metric for development choices. The early-stopping log later is a separate paper-policy exercise, not a training run generated by these steps.

Assume finite numeric values and nonnegative penalty. At the new v, data loss is 0.4, penalty 1.28 and total 1.68.

Choose penalty strength using a declared validation **prediction** metric, not by comparing total objectives with different lambdas as if they were the same measure.

Runnable separated-term calculation

```
def terms(v, penalty):
    data_loss = 2.5 * (v - 2) ** 2
    weight_cost = penalty * v * v
    gradient = 5 * (v - 2) + 2 * penalty * v
    return data_loss, weight_cost, data_loss + weight_cost, gradient

v = 1.5
penalty = 0.5
old = terms(v, penalty)
v = v - 0.1 * old[3]
print(round(v, 6))  # 1.6
print([round(x, 6) for x in terms(v, penalty)])
# [0.4, 1.28, 1.68, -0.4]
```

Connect visual values to their named term

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Separate a changed penalty from a changed stopping policy

> They act at different places in the experiment.

Current next-update task

- Start v = 1.6 with lambda = 0.5 and rate 0.1. Calculate both gradient contributions, new v and new **prediction** MSE.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured patience change

- Change: Change **patience** from 1 to2 under the same tie rule.

▼ **Check after predicting**

Prediction: Stop at checkpoint 4; still **restore** checkpoint 2.

Reason: Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

See the objective update and the policy replay as different mechanisms

> Keep each log's checkpoint numbering beside its values.

Complete pictured objective,log and separate scope

Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule.

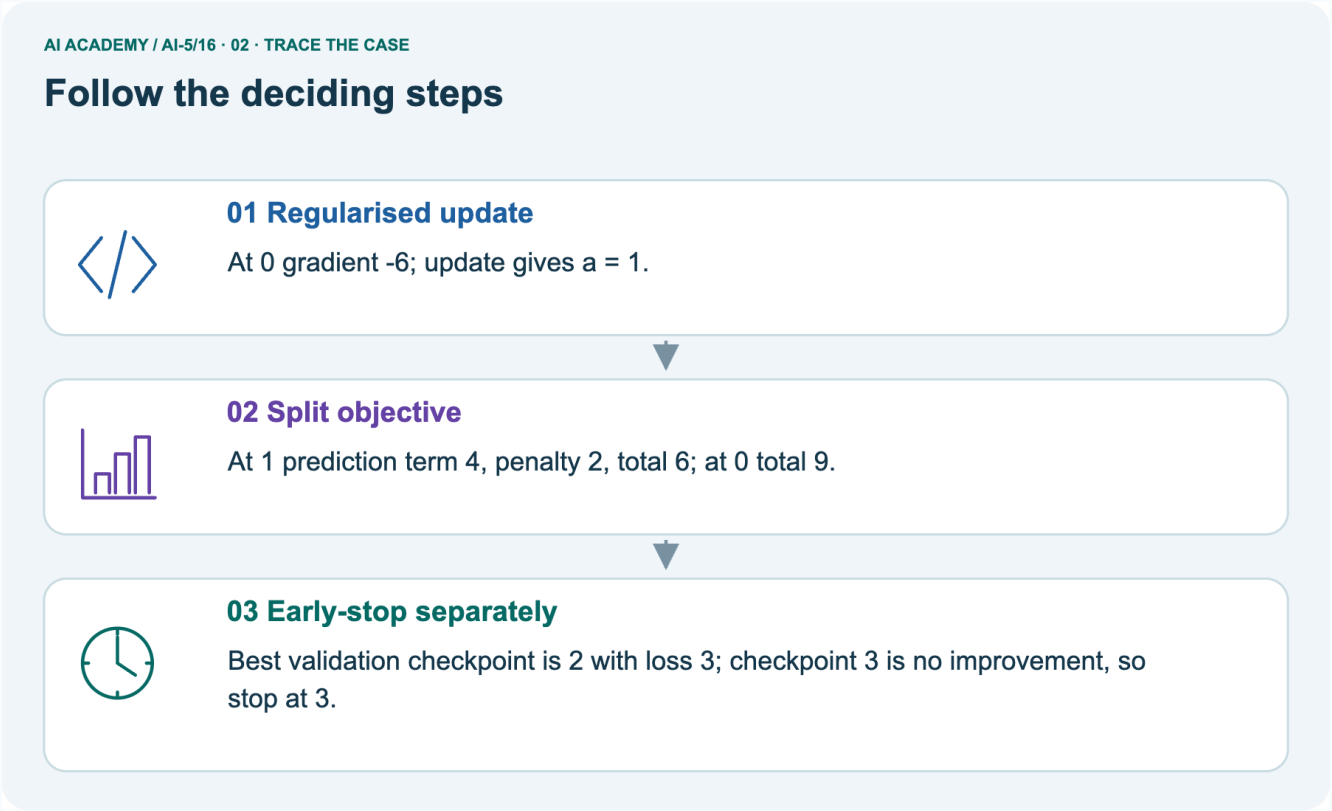
Toy objective $J(a)=(a-3)^2+2a^2$; start a = 0; gradient 6a-6; rate 1/6. Separate validation losses: [5, 3, 3, 4], **patience** 1, no ties as improvement.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture has two separate exercises. Its penalized objective $J(a)=(a-3)^2+2a^2$, gradient 6a−6, start 0, rate 1/6 gives a1:prediction term 4, penalty 2, total 6 instead of9. Its independent supplied validation log[5, 3, 3, 4]uses checkpoint numbers 1–4. With strict improvement and patience 1, checkpoint 2is best, checkpoint 3is the first tie/miss, so stop 3 and select saved 2.

With patience 2, stop 4 and still select 2. Those losses are supplied, not computed from this one update. The main stopping log uses epochs 0–7 instead.

Trace the penalty terms and supplied stopping choice



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Regularised update	At 0 gradient -6; update gives a = 1.
2. Split objective	At 1 prediction term 4, penalty 2, total 6; at 0 total 9.
3. Early-stop separately	Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

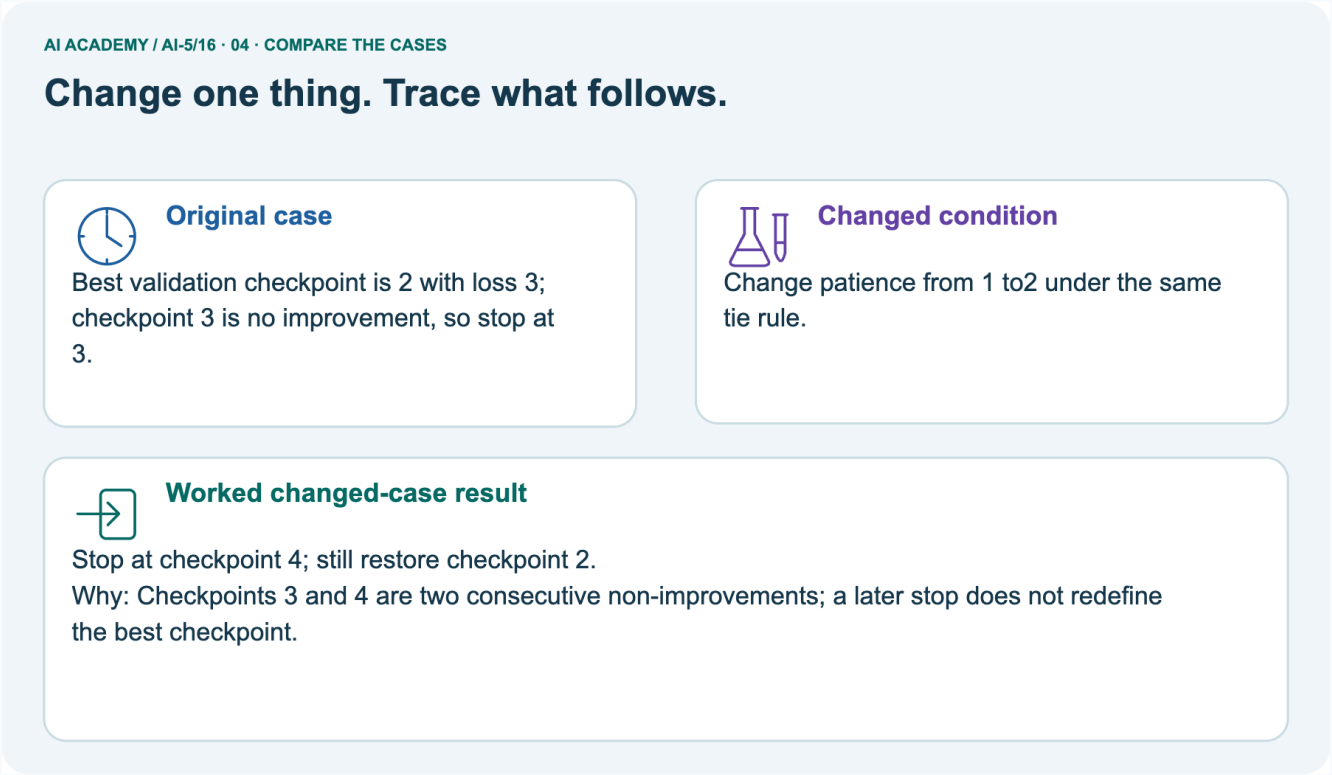
Regularisation adds a parameter penalty to the **training** objective. Here it discourages large a, so the objective balances **prediction** error against that penalty.

Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the **patience** allowance is used and **restore** the best checkpoint.

Predict the patience2policy

- Change one thing: Change **patience** from 1 to2 under the same tie rule. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain metric separation and tool limits

- **Regularisation** and early stopping are different tools; neither guarantees better performance in every dataset.

- Read the labelled picture: Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the main log’s miss counts

Patience two with strict improvement

EPOCH	VALIDATION MSE	DECISION
0 then 1 then 2	4 then 2 then 1.5	Save each improved state
3 then 4	1.6 then 1.4	Wait once then reset
5 then 6	1.45 then 1.46	Two misses so stop at 6

Initial checkpoint is epoch 0. Evaluate once after each epoch. A strict decrease is improvement; an exact tie is a miss. Reset the miss count after improvement and stop at two consecutive misses. Restore epoch 4, the best observed checkpoint.

At epochs 3, 4, 5 and 6, record the miss count after evaluation. Which epoch stops training, and which epoch supplies the restored parameters?

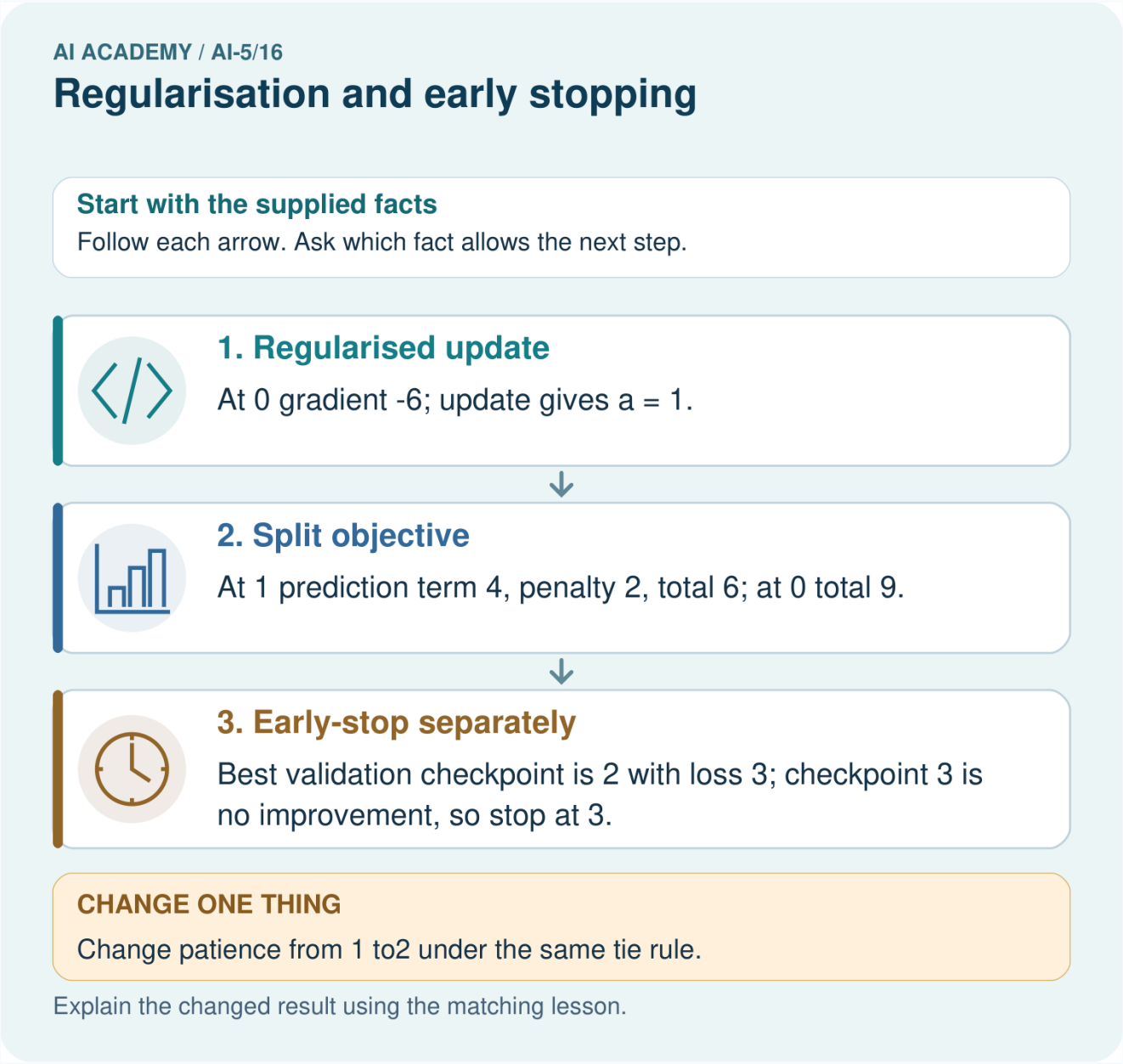
▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Regularisation and early stopping

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Stop when the rule fires,then restore the best observed model

> A later value in a paper log cannot rewrite what a live run observed.

Complete policy,finite budget and saved-state requirements

Early stopping limits the **training** duration using development **evidence**; it does not add a weight term to the objective. For this policy, initialise the best checkpoint to epoch 0 and the miss count to zero.

After each later evaluation, save and reset only on a strict improvement; otherwise increment the count. Stop as soon as the count reaches **patience**, then reload the best checkpoint.

Suppose a pre-recorded demonstration also lists epoch 7 loss 1.3. A live run following the rule has already stopped at 6 and does not observe 7. Do not use future values to rewrite the stopping decision.

The policy can miss a later improvement; **patience** balances waiting against a **training** budget and must be specified in advance.

Save actual model parameters with the checkpoint, not just its epoch number or loss. With a fixed maximum epoch budget, stop when either limit is reached and **restore** the best observed state.

An exact tie keeps the earlier checkpoint under our rule. Different minimum-improvement thresholds are possible, but would be different declared policies.

Every main-log decision,including unobserved future

Main supplied epoch · Validation MSE · Consecutive misses after evaluation · Best observed epoch · Policy action; patience 2

> **Main supplied epoch:** 0
>
> **Validation MSE:** 4
>
> **Consecutive misses after evaluation:** 0

Best observed epoch: 0

Policy action; patience 2: Save initial best

Main supplied epoch: 1

Validation MSE: 2

Consecutive misses after evaluation: 0

Best observed epoch: 1

Policy action; patience 2: Strict improvement:save/reset

Main supplied epoch: 2

Validation MSE: 1.5

Consecutive misses after evaluation: 0

Best observed epoch: 2

Policy action; patience 2: Strict improvement:save/reset

Main supplied epoch: 3

Validation MSE: 1.6

Consecutive misses after evaluation: 1

Best observed epoch: 2

Policy action; patience 2: Wait

Main supplied epoch: 4

Validation MSE: 1.4

Consecutive misses after evaluation: 0

Best observed epoch: 4

Policy action; patience 2: Strict improvement:save/reset

Main supplied epoch: 5

Validation MSE: 1.45

Consecutive misses after evaluation: 1

> **Best observed epoch:** 4
>
> **Policy action; patience 2:** Wait

> **Main supplied epoch:** 6
>
> **Validation MSE:** 1.46
>
> **Consecutive misses after evaluation:** 2
>
> **Best observed epoch:** 4
>
> **Policy action; patience 2:** Stop; select saved epoch 4

> **Main supplied epoch:** 7
>
> **Validation MSE:** 1.3
>
> **Consecutive misses after evaluation:** Not observed
>
> **Best observed epoch:** Not eligible
>
> **Policy action; patience 2:** Pre-recorded future value; live policy already stopped

Apply the full policy to the given log

- Validation MSE at epochs 0 to 7 is [4, 2, 1.5, 1.6, 1.4, 1.45, 1.46, 1.3]. Apply the lesson policy with **patience** 2. Give the stop epoch, **restore**d epoch and why the listed epoch 7 cannot change this decision.

> **State one repair and the checks you would rerun.**

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise two mechanisms with less help

> Explain which numerical quantity or decision changes.

Current task and response

- At v = 2 and lambda = 0.5 in the lesson, compute **prediction** MSE, penalty, total objective and total gradient. Does zero **training** MSE make this gradient zero?

At epochs 3, 4, 5 and 6, record the miss count after evaluation. Which epoch stops training, and which epoch supplies the restored parameters?

Explain your prediction to a partner before comparing results.

Supported W2record

Use workbook W2 for the connected example "**Regularisation** and early stopping". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Regularisation** and early stopping. Proposed explanation: “Zero **prediction** error means no regularised update”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured penalty/metric reasoning

- Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent new objective and log

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Trace the next update

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete penalty calculation and a separate policy replay

> The code makes the observed prefix explicit and stops before future values.

Current penalty task and original runnable terms helper

Start v = 1.6 with lambda = 0.5 and rate 0.1. Calculate both gradient contributions, new v and new **prediction** MSE.

```
def terms(v, penalty):
    data_loss = 2.5 * (v - 2) ** 2
    weight_cost = penalty * v * v
    gradient = 5 * (v - 2) + 2 * penalty * v
    return data_loss, weight_cost, data_loss + weight_cost, gradient

v = 1.5
penalty = 0.5
old = terms(v, penalty)
v = v - 0.1 * old[3]
print(round(v, 6))  # 1.6
print([round(x, 6) for x in terms(v, penalty)])
# [0.4, 1.28, 1.68, -0.4]
```

Complete runnable stopping-policy replay

```
from math import isfinite
```

```
def replay_stopping(losses, patience, first_epoch=0):
    """Replay supplied validation observations; no training or real files are loaded."""
    if not losses or type(patience) is not int or patience < 1:
        raise ValueError('Need observations and positive integer patience')
    best_index = 0
    misses = 0
    trace = []
    stop_index = len(losses) - 1  # finite supplied observation budget
    for i, value in enumerate(losses):
        if not isfinite(value) or value < 0:
            raise ValueError('Observed MSE must be finite and nonnegative')
        if i == 0:
            decision = 'save initial best'
        elif value < losses[best_index]:
            best_index, misses = i, 0
            decision = 'save improved best'
        else:
            misses += 1
            decision = 'stop; select saved best' if misses >= patience else 'wait'
        trace.append((first_epoch + i, value, misses, first_epoch + best_index, decision))
        if misses >= patience:
            stop_index = i
            break  # do not inspect or select from later supplied observations
    return first_epoch + stop_index, first_epoch + best_index, trace

# Separate supplied paper logs; values are not generated by the penalty code.
main = [4, 2, 1.5, 1.6, 1.4, 1.45, 1.46, 1.3]
stop, best, observed = replay_stopping(main, patience=2)
print('main stop / selected saved epoch', stop, best)
print('observed trace: epoch, loss, misses, best epoch, action')
for row in observed:
    print(row)
print('picture patience 1, checkpoints numbered from 1:', replay_stopping([5,3,3,4], 1, 1)
[:2])
print('picture patience 2, checkpoints numbered from 1:', replay_stopping([5,3,3,4], 2, 1)
[:2])
print('Policy decisions replayed. Actual saved model files are not supplied or restored by
this program.')
```

Normal,boundary,missing-snapshot and future-evidence probes

Complete objective or stopping probe	Expected answer / action
Normal penalty update v1.6, lambda.5, rate.1	Data gradient−2, penalty 1.6, total−.4, newv 1.64, data MSE.324
Penalty boundary lambda 0	Jequals prediction MSE; penalty gradient 0
Main stopping logpatience 2	Stop6, select saved 4; misses at3/4/5/6 are1/0/1/2
Picture logcheckpointnumbers 1–4, patience 1	Stop3, select 2; ties are not improvement
No misses before finite log ends	End at supplied budget; select best observed state
Replay empty log or patience 0	ValueError:Need observations and positive integer patience
Missing actual model snapshot	Replay can select an epoch only; request/load its actual saved state before claiming restoration
Future epoch 7 loss1.3 after main stop 6	Not inspected or used to choose; do not substitute it for saved 4
Choose lambda/patience with final-test results	Settings become test-informed development choices; freeze a new policy and reserve fresh final evidence

Predict,calculate,replay and retain actual status

1. First trace the complete separated-term update. Then predict the main and pictured stopping decisions before running or tracing the standalone policy program. Save the observed miss-count ledger and the selected epoch. The program replays supplied losses; it neither trains a network nor loads/restores actual model files, which are not provided with these logs.

   For your project, save and reload actual parameters and any required model state, then verify that restored version. Keep the independent new objective/log for a first attempt before feedback. Neither penalty nor stopping guarantees improved independent performance.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: a new penalty objective and a separate stopping log

> Use its supplied gradient and zero-based epochs;keep the two tasks distinct.

Complete new independent objective and policy

For a new objective J(a) = (a - 2) squared + a squared, gradient 4a - 4, start a = 0 and take one step at rate 0.25. Give new a, **prediction** error term, penalty and total J.

Separately apply **patience** 2 with strict improvement to validation values [3, 2, 2, 2, 1] at epochs 0 to 4; report the stop and **restore**d epochs.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer both rules independently

Attempt this without hints

For a new objective J(a) = (a - 2) squared + a squared, gradient 4a - 4, start a = 0 and take one step at rate 0.25. Give new a, **prediction** error term, penalty and total J.

Separately apply **patience** 2 with strict improvement to validation values [3, 2, 2, 2, 1] at epochs 0 to 4; report the stop and **restore**d epochs.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the objective and restoration policy with the project

> A selected epoch number cannot replace the actual saved model state.

Project artifact and evidence table

Add a labelled **regularisation** and early stopping **evidence** sheet to Image-pattern Network Investigation. Show how you can separate a **weight penalty** from **prediction** error.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with separate loss terms and a justified saved-state choice

> Name what was computed,selected and actually restored.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Separate a **weight penalty** from **prediction** error

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Regularisation	A constraint or training procedure intended to limit fitting choices and improve generalisation
Weight penalty	An added objective term that discourages specified large parameter values
Patience	The allowed run of consecutive non-improving evaluations before stopping under a stated rule
Restore	Reload the parameters saved at the selected checkpoint

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- At v = 2 and lambda = 0.5 in the lesson, compute **prediction** MSE, penalty, total objective and total gradient. Does zero **training** MSE make this gradient zero?

- Validation MSE at epochs 0 to 7 is [4, 2, 1.5, 1.6, 1.4, 1.45, 1.46, 1.3]. Apply the lesson policy with **patience** 2. Give the stop epoch, **restore**d epoch and why the listed epoch 7 cannot change this decision.

- A team compares lambda 0 and lambda 0.5 by choosing whichever has the smaller penalised **training** objective. Explain why this does not establish the better **prediction** model and name a suitable development comparison.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent penalty objective and zero-based stopping log for the first individual attempt.

Current explanation and response

- Weight penalties and early stopping can be used together, but they act differently.

  Record which weights receive penalties, the objective convention, lambda, learning rate, checkpoint frequency, **patience**, tie rule and maximum budget.

  For this classroom model there are no extra learned pre**process**ing or running statistics; larger models may require more state to reproduce evaluation.

- Select these settings using validation, freeze the complete procedure, and reserve untouched final test data for reporting. Neither **regularisation** nor stopping makes a weak validation set representative.

  Next week studies initialisation and reproducibility.

- A team compares lambda 0 and lambda 0.5 by choosing whichever has the smaller penalised **training** objective. Explain why this does not establish the better **prediction** model and name a suitable development comparison.

A team stops at epoch 6, uses its final parameters instead of the saved best state, and chooses patience by looking at final test losses. Identify the two mistakes and explain how to repair the experiment.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Patience two with strict improvement

EPOCH	VALIDATION MSE	DECISION
0 then 1 then 2	4 then 2 then 1.5	Save each improved state
3 then 4	1.6 then 1.4	Wait once then reset
5 then 6	1.45 then 1.46	Two misses so stop at 6

Initial checkpoint is epoch 0. Evaluate once after each epoch. A strict decrease is improvement; an exact tie is a miss. Reset the miss count after improvement and stop at two consecutive misses. Restore epoch 4, the best observed checkpoint.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule.

By the end, I can	How I will show it
Separate a weight penalty from prediction error	Explain the deciding step, show a trace or test, and state one limit.
Trace a regularised update under a stated objective	Explain the deciding step, show a trace or test, and state one limit.
Apply a stopping rule and restore the best observed checkpoint	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule.
- Which labelled step matches this detail? Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.
- 1. Split objective
- 2. Early-stop separately
- 3. Regularised update
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Early-stop separately Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

Regularisation adds a parameter penalty to the **training** objective. Here it discourages large a, so the objective balances **prediction** error against that penalty.

Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the **patience** allowance is used and **restore** the best checkpoint.

Changed-case question and optional hint

- Changed-case practice: Change **patience** from 1 to2 under the same tie rule. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Stop at checkpoint 4; still **restore** checkpoint 2.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Stop at checkpoint 4; still **restore** checkpoint 2. Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Repair and retain the record

Review **Training** and validation curves (ai-5/15). Goal: Compute comparable **training** and validation losses at saved checkpoints. Short practice: Can the original epoch 3 **training** loss 1 justify claiming best generalisation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Batching and epochs (ai-5/13). Goal: Distinguish an example a batch an update and an epoch. Short practice: Why do the single-example orders finish at different weights?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

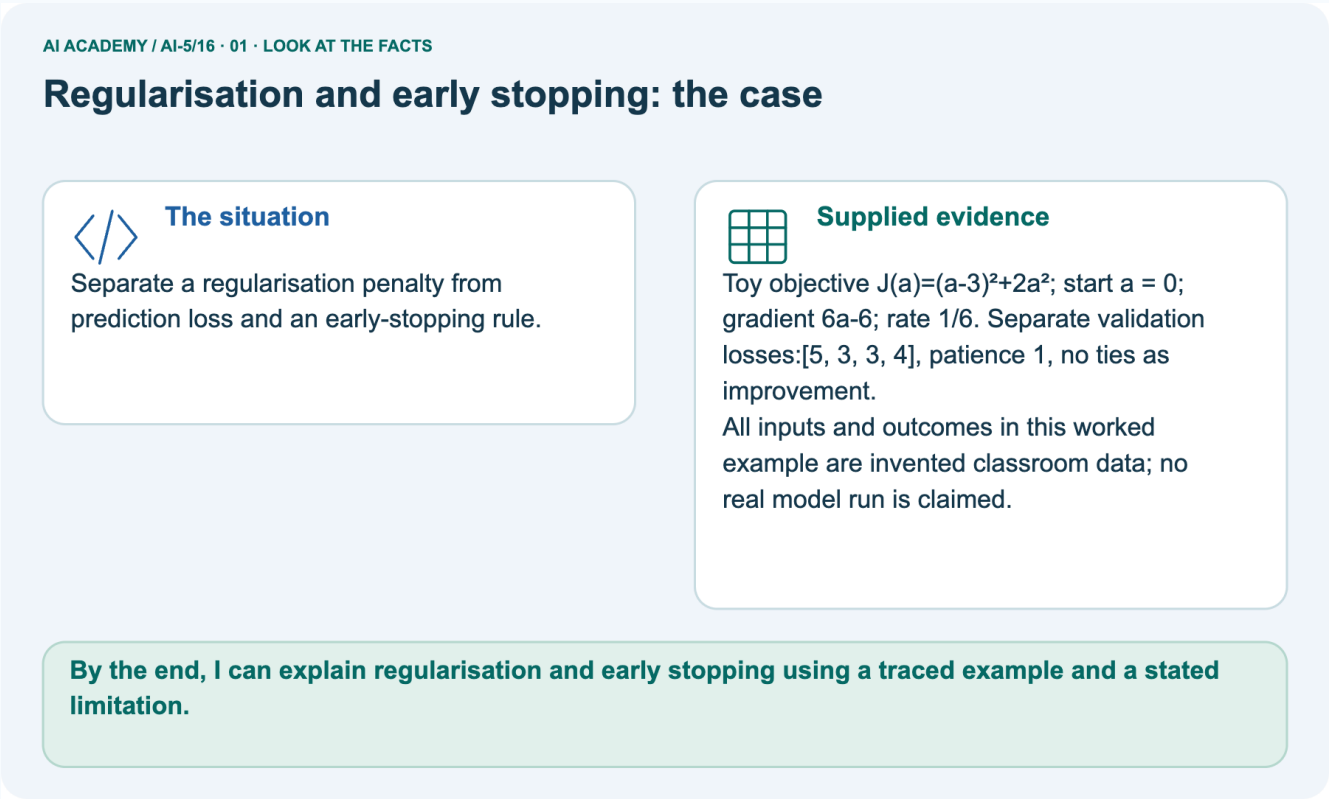
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Regularisation and early stopping: the case



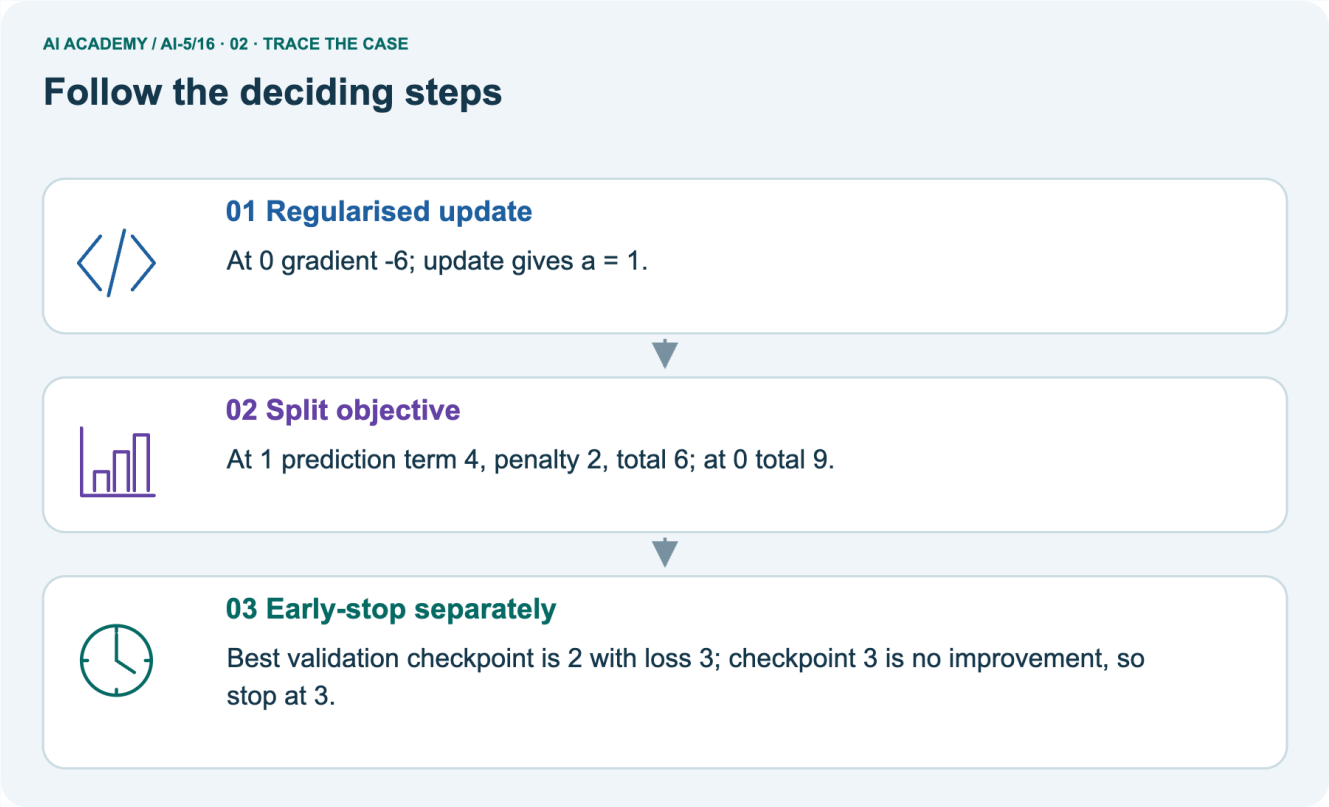
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule. Supplied facts: Toy objective J(a)=(a-3)²+2a²; start a = 0; gradient 6a-6; rate 1/6.

Separate validation losses:[5, 3, 3, 4], **patience** 1, no ties as improvement. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Regularised update: At 0 gradient -6; update gives a = 1. → Split objective: At 1 **prediction** term 4, penalty 2, total 6; at 0 total 9. → Early-stop separately: Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

Point to each numbered stage and explain which recorded **input** it uses.

03. Patience two with strict improvement

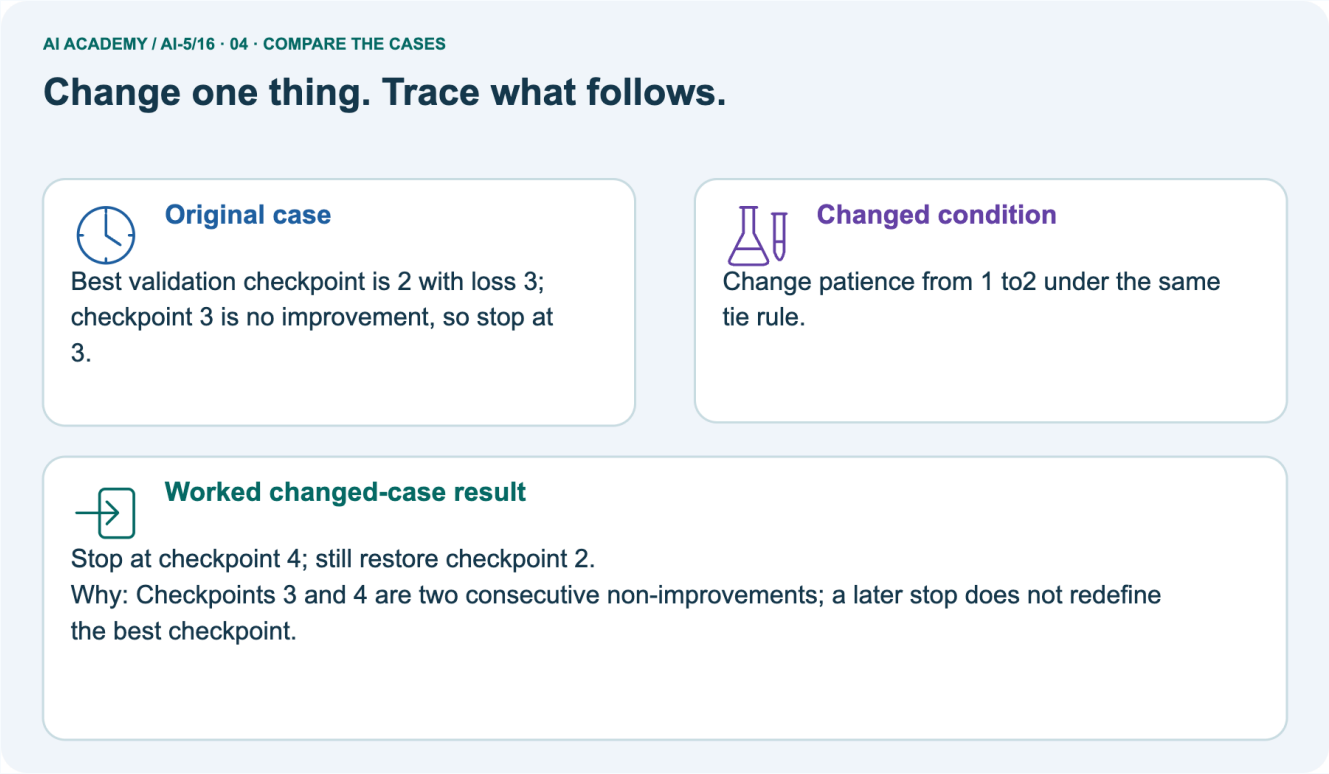


Follow the labelled picture with the matching explanation. Open the vector diagram

Patience two with strict improvement. 0 then 1 then 2 | 4 then 2 then 1.5 | Save each improved state. 3 then 4 | 1.6 then 1.4 | Wait once then reset. 5 then 6 | 1.45 then 1.46 | Two misses so stop at 6

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



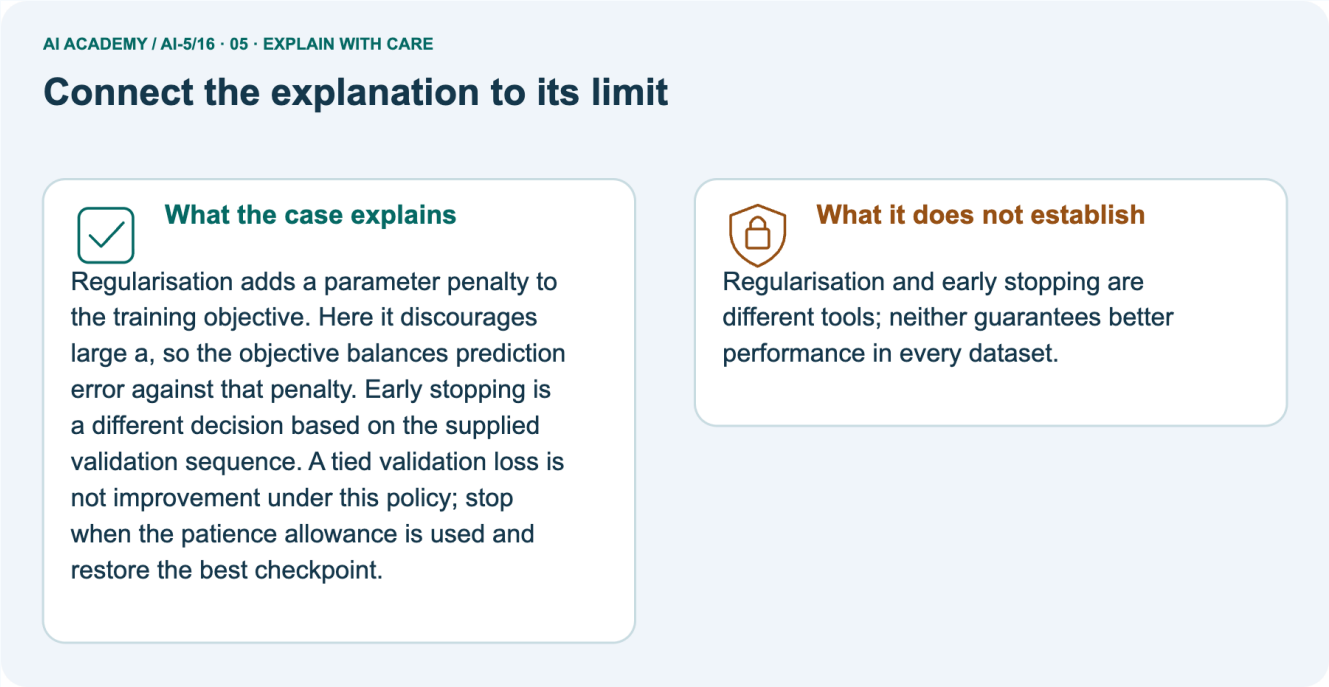
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3. Changed condition: Change **patience** from 1 to2 under the same tie rule.

Expected result: Stop at checkpoint 4; still **restore** checkpoint 2. Reason: Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

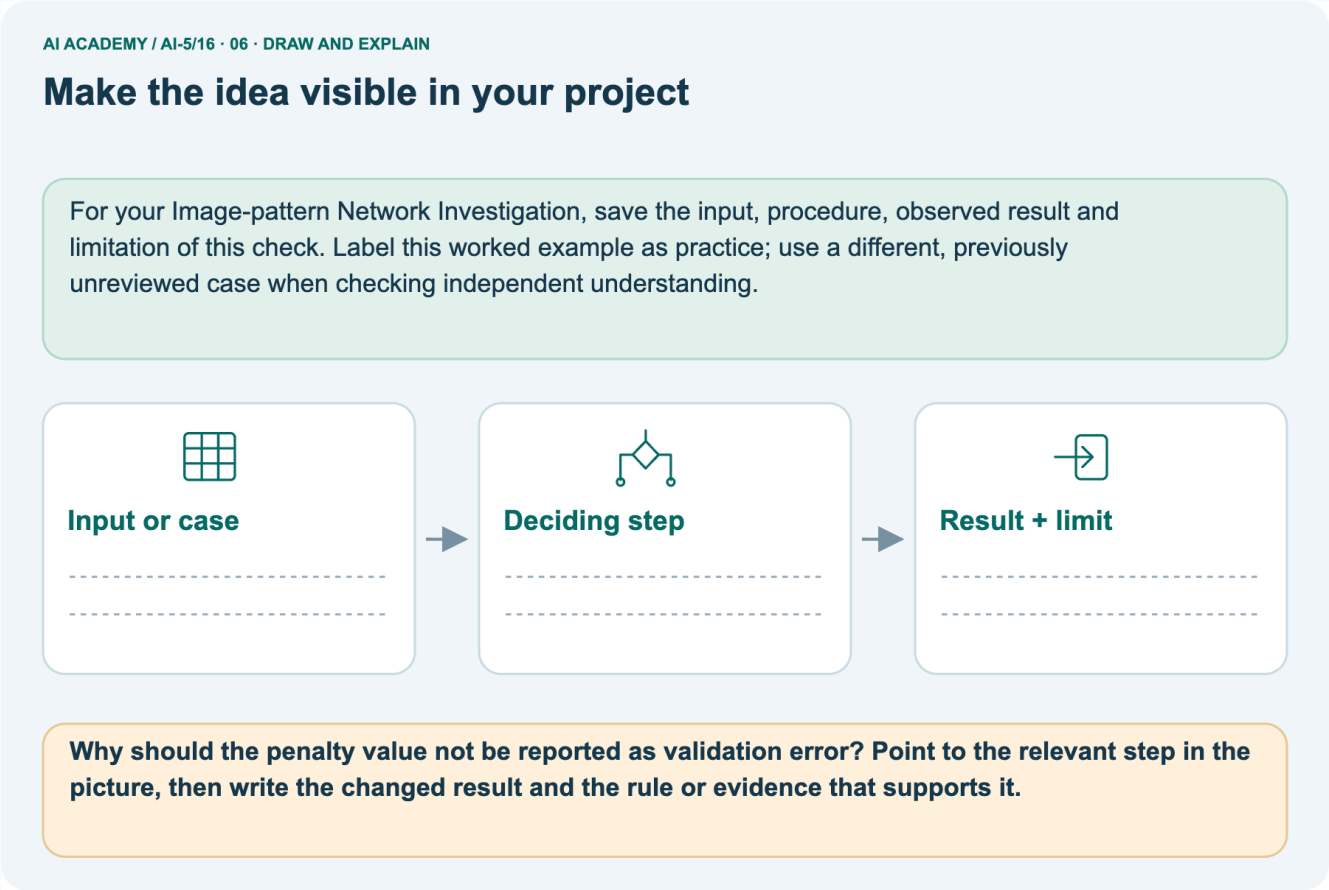
Regularisation adds a parameter penalty to the **training** objective. Here it discourages large a, so the objective balances **prediction** error against that penalty.

Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the **patience** allowance is used and **restore** the best checkpoint.

Limit: **Regularisation** and early stopping are different tools; neither guarantees better performance in every dataset.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Regularisation and early stopping: the case

02. Follow the deciding steps

03. Patience two with strict improvement

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.